

DAILY DRILLING REPORT (DDR) - RIG 04

SYNTHETIC DEMONSTRATION DOCUMENT - NOT REAL OPERATIONAL DATA (SIH PROTOTYPE)

WELL IDENTIFIER: NWIS-W002

OPERATIONAL DATE: 2024-04-18

FORMATION: Demo-Barail

MEASURED DEPTH: 2980.0 m

EVENT TYPE: MUD_LOSS

SEVERITY LEVEL: HIGH

OPERATIONAL SUMMARY

Drilled 8-1/2 inch hole section from 2940m to 2980m in Demo-Barail formation.
At 2980m measured depth, severe dynamic mud loss was observed at surface.
Loss rate abruptly exceeded 45 bbls/hr (7.15 m3/hr). Standpipe pressure dropped from 2350 psi to 1920 psi.

ROOT CAUSE ANALYSIS

Penetrated depleted pore pressure sandstone reservoir with subnormal pressure gradient (~0.38 psi/ft).
Excess hydrostatic head from 1.34 SG active drilling mud exceeded formation breakdown pressure,
inducing fracture opening in permeable Barail sand facies.

CORRECTIVE MITIGATION

1. Immediately reduced circulation rate from 2400 L/min to 1600 L/min.
2. Mixed and spotted 50 bbls coarse calcium carbonate (CaCO3) Loss Circulation Material (LCM) pill.
3. Reduced active mud weight to 1.22 SG using low-solids water-based polymer mud.
4. Staged resumption of drilling; loss rate successfully arrested to <2 bbls/hr seepage.